

Baseline noise in mass spectrometer measurements

Client project report

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492,000

Baseline readings

82,000

Readings per dwell time

6

Dwell-time settings

Summary

Background noise should be estimated using the measurement time of each instrument channel. The instrument records a small electrical background even when no sample signal is present. Our analysis of 492,000 baseline readings found that both the typical noise level and its variation changed substantially with measurement time. A correction based on one timing setting may therefore misrepresent the background in another channel.

We recommend developing and testing a correction that matches each channel's measurement time. The project provides a statistical foundation for this work. It has not yet shown that the proposed approach improves real sample measurements or the rock ages calculated from them.

Project Question

How does instrument background noise behave, and how should that knowledge guide more reliable estimates of the true sample signal? We focused on whether noise follows a consistent pattern and how that pattern changes with **dwell time**, the time the instrument spends collecting one reading.

Findings

How background noise changes with measurement time

01 Longer dwell times had lower and less variable background noise.

Across the tested settings, the typical noise level, measured by the median, was about 82% lower at 4.096 seconds than at 0.128 seconds. The longest setting takes 32 times as long per reading, so this benefit comes with a measurement-time tradeoff.

02 Noise was unevenly distributed.

Most readings were relatively small, with occasional much larger values. These larger readings pulled the average above the median at every setting. A single average does not describe the full range of background noise.

03 The baseline channels did not closely track one another.

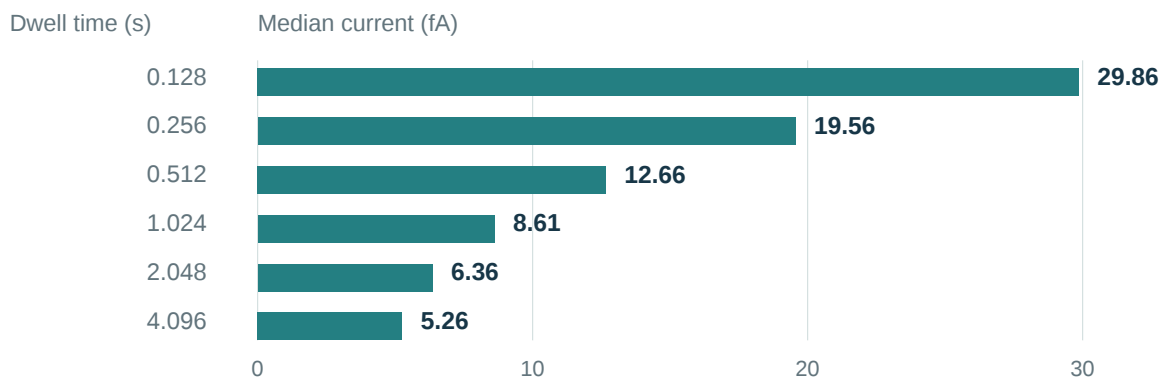
Baseline readings at different dwell times generally did not rise and fall together within the same cycle. This limits the case for subtracting one live baseline value from every channel.

04 A flexible model captured much of the observed pattern.

The fitted model allows noise behavior to change with dwell time. Its remaining mismatches, including those at the extremes, make it a candidate for further testing rather than a finished correction method.

Typical background noise by measurement time

Median background current at the six tested dwell times



The median is the middle reading: half the observations are below it and half above it. Bars show the six tested settings; they do not establish behavior at untested times.

Source: [3], saved dwell-time summaries. All values are in femtoamperes.

Suggestions and limitations

Practical next steps and boundaries on interpretation

Suggestions

- 1. Test a correction matched to each channel's dwell time.** Use baseline measurements taken at the corresponding setting. Start with the observed median as a simple comparison method, and assess whether the fitted noise model improves on it.
- 2. Keep the live baseline channel for monitoring.** Use it to flag changes in instrument behavior. Test whether it adds useful information for a particular channel before using it as a direct correction for that channel.
- 3. Validate with blanks and samples with known signals.** Compare the current method, a median correction matched to dwell time, and the fitted model on independent runs. Measure recovery of known signals, remaining error, and uncertainty, especially for weak signals. Check performance separately by instrument and laboratory.
- 4. Build uncertainty into the final signal estimate.** Develop a method that keeps estimated true signals at or above zero and reports a plausible range. Simple baseline subtraction can still produce negative estimates. Signals close to the background level need a clear rule for reporting uncertainty or a detection limit.
- 5. Choose longer dwell times selectively.** Consider them where weak signals need greater precision, while checking the effect on total measurement time. Repeat baseline checks after changes to instrument settings, maintenance, or operating conditions.

Limitations

The study measured background noise, not correction performance. It did not demonstrate improved accuracy for unknown samples, isotope ratios, or rock-age estimates. The practical benefits remain to be measured.

The evidence covers a defined set of conditions. The analysis concerns three Prisma Pro instruments at Glasgow, Salzburg, and Wuhan, using six dwell times from 0.128 to 4.096 seconds. A separate Salzburg batch was labeled Nineamu. Results should be checked before transfer to other instruments, settings, or periods of operation.

The model is an approximation. Laboratory and batch differences remain relevant, and the fitted model does not explicitly adjust for them. Only one fitted statistical model is reported, so it has not been established as the best available choice. Its most unusual predicted readings need particular scrutiny.

Data handling and repeated measurements affect interpretation. Cleaning relied on assumptions about automatic baseline subtraction and timestamp resets. Three zero readings were excluded from logarithmic modeling. Many readings came from the same runs; the large observation count alone does not establish long-term stability or independence.

Supporting statistics

The cleaned dataset contains **492,000 readings**, with **82,000 at each dwell time**. The table and chart summarize those readings, including the three zeros. They describe measured background noise, not corrected sample signals. [2, 3]

Noise level and spread at each setting

Current values are in **femtoamperes (fA)**, where $1 \text{ fA} = 10^{-15} \text{ amperes}$. Smaller values represent lower background current.

Dwell time (seconds)	Median (fA)	Mean (fA)	Standard deviation (fA)	95th percentile (fA)
0.128	29.86	39.28	34.87	108.40
0.256	19.56	24.90	21.39	66.62
0.512	12.66	15.65	12.92	40.81
1.024	8.61	10.25	7.97	25.57
2.048	6.36	7.16	5.13	16.65
4.096	5.26	5.64	3.70	12.25

Sources: [2], summary statistics; [3], saved numerical summaries. Values are rounded. Shading highlights the shortest and longest dwell times.

How to read the statistics

Mean is the arithmetic average. **Standard deviation** measures spread; a smaller value means less variation. The **95th percentile** is the level below which approximately 95% of the baseline readings fall. It is not a validated sample detection limit.

From the shortest to the longest setting, the median fell by **82.4%**, the mean by **85.6%**, and the standard deviation by **89.4%**. These are descriptive comparisons across settings, not measured gains in sample accuracy.

For context, the median background at the live baseline timing of 0.512 seconds was about **2.4 times** the median at 4.096 seconds. This illustrates why a timing mismatch matters; it is not a rule for scaling individual live readings.

Methods and technical notes

How the analysis was carried out

Baseline Characterization Analysis (BCA) runs measure the instrument at a mass setting where sample signal is assumed to be absent. The preprocessing workflow parsed the measurement files, reversed suspected automatic baseline subtraction, and retained the final uninterrupted segment after timestamp resets. The cleaned data had no negative readings. Removing three zeros for the log transform left **491,997 positive observations** for modeling. [1-4]

A **log-skew-t model** was fitted to the natural logarithm of current. It uses seven fitted parameters: two each for how location, scale, and tail behavior change with log dwell time, plus one shared skewness parameter. In plain terms, it describes how the position, width, and shape of the noise distribution change with measurement time. Histograms and kernel density estimates, which are smoothed summaries of observed readings, provided comparison curves. [4]

What the diagnostic results show

Cycle-aligned Spearman rank correlations averaged **0.0073**, with pairs ranging from **-0.0948 to 0.1335**. Values near zero indicate weak rank association. This supports caution about transferring live baseline readings across dwell times; it does not prove independence or rule out changes within individual runs. [3]

The saved fit reports **AIC 1,351,186.11** and **BIC 1,351,263.86**. These are relative model-comparison scores, not accuracy percentages or pass marks. Alternative models must use comparable observations and likelihood scales for these scores to be compared meaningfully. No independent test of correction performance is reported. [2, 4]

Checks needed before implementation

The technical report lists shared skewness as **0.0592**, while the saved model output gives **-3.8056**. The accompanying claim of mild positive skewness on the log scale therefore needs reconciliation. Similarly, a narrative note in the assumptions notebook gives a different correlation average; the matrix and saved summary support **0.0073**, used above. These discrepancies do not change the observed dwell-time trend, but model settings must be checked before reuse. [2-4]

Use medians or validated quantiles when developing an initial correction. A positive noise model does not itself ensure a nonnegative corrected signal. Confirm the handling of near-zero signals, check unusually large predictions, and test uncertainty estimates on runs excluded from model fitting. Laboratory and batch effects should be assessed before adopting a pooled calibration.

Project sources

[1] Project Background.docx. Instrument context, acquisition settings, and project objectives.

[2] Melendez, Austin, and Harmen Hundal. *Identifying the Distribution of Baseline Noise in Mass Spectrometer Measurements*. Technical Report.docx, May 19, 2026. Sections 2-6 and appendices.

[3] 02-assumptions.ipynb. Saved dwell-time summaries, laboratory summaries, correlation matrix, and final analysis summary.

[4] 01-preprocessing.ipynb and 03-modeling.ipynb. Processing rules, model specification, saved fit, and diagnostic plots.